

Light Sources and Rays

Light enables us to **see** things.

Light is made by **light sources**.

We see a light source if its light ends up in **our eyes**.

1 Is it a light source? How did you decide?

(a) A TV screen,

(d) the Sun,

(b) a desk,

(e) a sheet of paper,

(c) a mobile phone screen,

(f) a poster on a wall.

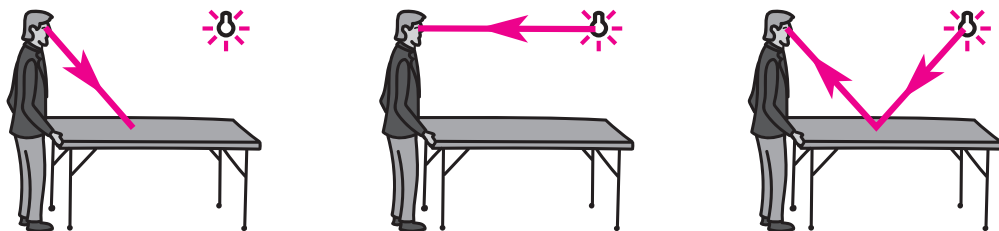
A cup is **not** a light source. We can see it when

- light from a light source
- travels to the cup,
- then **bounces off** the cup.
- The light then travels from the cup to **our eyes**.

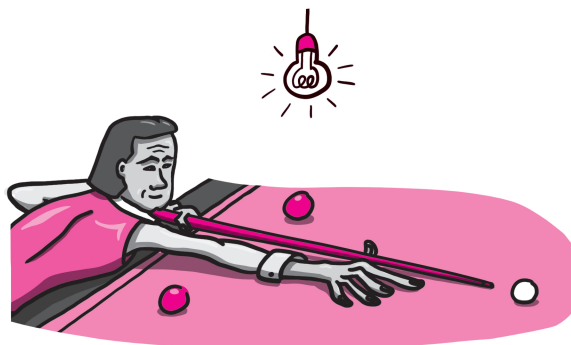
When light bounces off something, we say it **reflects**.

Light travels through the air in a **straight line**.

2 A person can see a table in a room with a light bulb. Which of the following diagrams is correct? The lines with arrows show the light travelling.



3 Draw lines with arrows to show how a snooker player is able to see a ball.



The straight lines with arrows on our diagrams are called **rays**.

- 4 A lifeguard on the beach sees a swimmer in need of help. Complete the sentences using the words **reflects**, **lifeguard**, **swimmer**, **Sun**, **eyes**. You can use a word more than once.

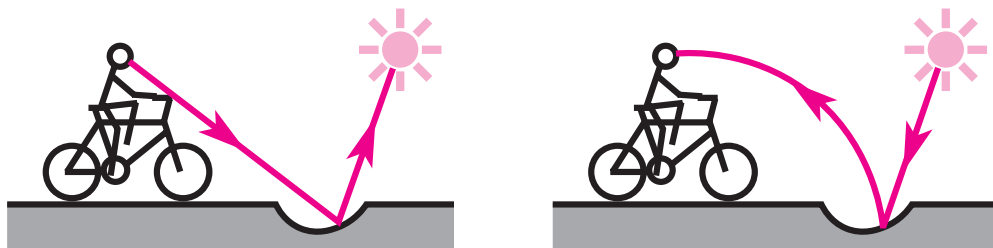
Light from the _____ travels to the _____ and _____ off them.

Light then travels from the _____ to the _____ and ends up in their _____.

- 5 What is wrong with these diagrams showing a person looking at their phone?



- 6 What is wrong with these diagrams showing a cyclist looking at a hole in the road?

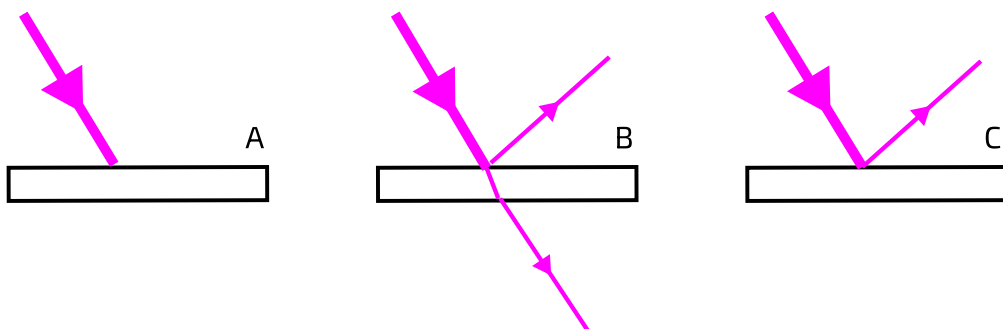


Light can not pass through an **opaque** material.

When light reaches an opaque object, there is **reflection** and **absorption**.

The **energy** from the light that is absorbed causes the material to warm up.
This increases the **thermal** energy store of the material.

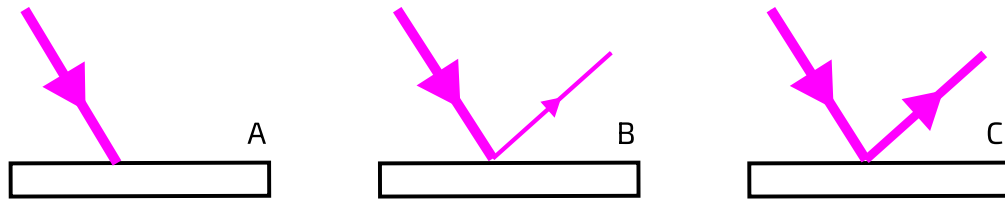
- 7 Which of the diagrams below show light reaching an opaque material?



8 In these diagrams, the thickness of the line shows the brightness of the light.

(a) Which of these diagrams show reflection?

(b) Which of these diagrams show absorption?



9 Complete the sentences using the words **reflected, absorbed, some, no, warmer, cooler, slowly, quickly, bright**. You can use a word more than once. Some words are not used

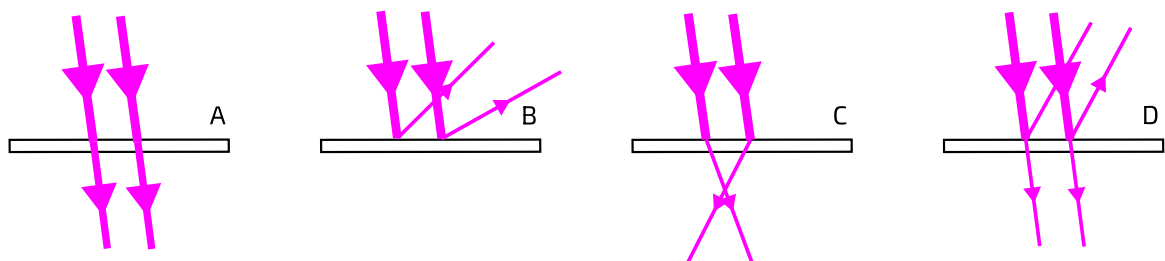
My desk is opaque and painted a pale colour. When sunlight shines on it, _____ light goes through it. I can see the desk, because most of the light is _____ by the surface. After the desk has been in sunlight for an hour, it feels much _____. This means that some sunlight was _____.

I paint my desk a darker colour. Less light will now be _____ off it, and so it won't look as _____. The desk will now warm up more _____ in the sunlight because more light is now being _____.

Light can go through a thin white sheet. Light can also go through a glass window. You can see what is on the other side of a glass window. You can't see through the thin white sheet clearly because the rays of light get **scattered (diffuse transmission)**.

The glass window is **transparent**. The thin white sheet is **translucent**.

10 Match the diagrams to the words **opaque, translucent** and **transparent**.



11 Do we need an opaque, transparent or translucent material? Fill in the table.

Description	Opaque, transparent or translucent?
The lens in a camera	
The window in a bathroom	
Bedroom curtain if you need to sleep during the day	